DISCRETE STRUCTURES

# CMSC250

## Chapter 25: Relations and Their Properties

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Ordered pairs, relation properties, graphs, and closures



# Contents

**1 Relations are sets of ordered pairs** **3**

**2 Directed graphs of relations** **3**

**3 Reflexive and irreflexive** **3**

**4 Symmetric, antisymmetric, and asymmetric** **3**

**5 Transitive relations** **4**

**6 Relation matrices** **4**

**7 Inverse relations and composition** **5**

**8 Closures of relations** **5**

**9 Practice questions and solutions** **5**

**10 Common mistakes** **6**

# 1 Relations are sets of ordered pairs

A *binary relation* $R$ from $A$ to $B$ is any subset of $A \times B$. A relation *on* $A$ is a subset of $A \times A$. We write $aRb$ to mean $(a, b) \in R$. Unlike a function, a relation may connect one input to several outputs, or none.

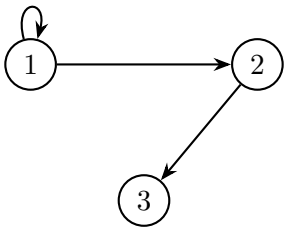
### Worked example: read a relation

**Question.** Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 3)\}$. Which statements are true?

**Solution.** $1R2$ and $2R3$ are true. $2R1$ and $3R3$ are false. The fact that $1R2$ holds does not imply $2R1$.

# 2 Directed graphs of relations

Represent each element of $A$ as a vertex and draw an arrow $a \to b$ exactly when $aRb$. A pair $(a, a)$ is a loop.




# 3 Reflexive and irreflexive

A relation $R$ on $A$ is *reflexive* if $\forall a \in A,\ aRa$. Every vertex must have a loop. It is *irreflexive* if $\forall a \in A,\ \sim (aRa)$. No vertex may have a loop. A relation may be neither reflexive nor irreflexive.

### Worked example: reflexivity of divisibility

**Question.** Prove that divisibility on the positive integers is reflexive.

**Proof.** Let $a$ be an arbitrary positive integer. Since $a = a \cdot 1$ and $1 \in \mathbb{Z}$, the definition of divisibility gives $a \mid a$. As $a$ was arbitrary, divisibility is reflexive. ■

# 4 Symmetric, antisymmetric, and asymmetric

A relation is *symmetric* if $aRb \Rightarrow bRa$ for all $a, b$. It is *antisymmetric* if $(aRb \wedge bRa) \Rightarrow a = b$. It is *asymmetric* if $aRb \Rightarrow \sim (bRa)$.

**Key idea.** Antisymmetric does **not** mean "not symmetric." A relation can be both symmetric and antisymmetric (for example, equality). Antisymmetry only prohibits arrows in both directions between *distinct* vertices.

### Worked example: antisymmetry of subset inclusion

**Question.** Prove $\subseteq$ on $\mathcal{P}(A)$ is antisymmetric.

**Proof.** Let $S, T \subseteq A$ be arbitrary. Assume $S \subseteq T$ and $T \subseteq S$. By the definition of set equality, the two inclusions imply $S = T$. Therefore $\subseteq$ is antisymmetric. ■

### Worked example: symmetry is not transitivity

**Question.** Let $A = \{1, 2\}$ and $R = \{(1, 2), (2, 1)\}$. Is $R$ symmetric? Transitive?

**Solution.** It is symmetric because both arrows occur. It is not transitive: $1R2$ and $2R1$ hold, but $1R1$ does not. A single missing required pair disproves transitivity.

## 5 Transitive relations

A relation is *transitive* if $\forall a, b, c \in A,\ (aRb \wedge bRc) \Rightarrow aRc$. In a directed graph, every two-step directed path requires a direct arrow from its start to its end.

### Worked example: transitivity of divisibility

**Question.** Prove divisibility on the positive integers is transitive.

**Proof.** Let $a, b, c$ be arbitrary positive integers. Assume $a \mid b$ and $b \mid c$. By definition, there exist integers $k, \ell$ such that $b = ak$ and $c = b\ell$. Substituting gives $c = (ak)\ell = a(k\ell)$. Since integers are closed under multiplication, $k\ell \in \mathbb{Z}$. Thus $a \mid c$. Therefore divisibility is transitive. ■

## 6 Relation matrices

For a finite ordered set $A = \{a_1, \ldots, a_n\}$, the *relation matrix* $M_R$ has entry 1 in row $i$, column $j$ if $a_i R a_j$, and 0 otherwise.

### Worked example: matrix representation

**Question.** Find the matrix of $R = \{(1, 1), (1, 2), (2, 3)\}$ on $A = \{1, 2, 3\}$.

**Solution.** With rows and columns ordered $1, 2, 3$,

$$M_R = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Reflexivity requires all diagonal entries to be 1; symmetry requires $M_R = M_R^{\mathsf{T}}$.

## 7 Inverse relations and composition

For $R \subseteq A \times B$, its inverse relation is $R^{-1} = \{(b, a) : (a, b) \in R\}$. If $R \subseteq A \times B$ and $S \subseteq B \times C$, then

$$S \circ R = \{(a, c) : \exists b \in B \ (aRb \wedge bSc)\}.$$

### Worked example: compose two relations

**Question.** Let $R = \{(1, a), (2, b)\}$ and $S = \{(a, x), (a, y), (b, y)\}$. Find $S \circ R$.

**Solution.** From 1 we can reach $a$, then $x$ or $y$. From 2 we can reach $b$, then $y$. Thus $S \circ R = \{(1, x), (1, y), (2, y)\}$.

## 8 Closures of relations

The *reflexive closure* adds every missing $(a, a)$. The *symmetric closure* adds reversed pairs; it is $R \cup R^{-1}$. The *transitive closure* adds all pairs needed to make the relation transitive, corresponding to reachability by directed paths.

### Worked example: transitive closure

**Question.** On $A = \{1, 2, 3\}$, let $R = \{(1, 2), (2, 3)\}$. Find its transitive closure.

**Solution.** The path $1 \to 2 \to 3$ forces $(1, 3)$. No other new paths arise. Hence the transitive closure is $\{(1, 2), (2, 3), (1, 3)\}$.

## 9 Practice questions and solutions

**Question.** 1. Is the relation $aRb \iff a - b$ is even on $\mathbb{Z}$ reflexive, symmetric, and transitive?

**Proof.** Reflexive: $a - a = 0$ is even. Symmetric: if $a - b = 2k$ then $b - a = -2k$ is even. Transitive: if $a - b = 2k$ and $b - c = 2\ell$, then $a - c = 2(k + \ell)$ is even by closure of integers under addition. Thus all three properties hold. ■**Question.** 2. Is $<$ on $\mathbb{Z}$ reflexive, symmetric, antisymmetric, or transitive?

**Solution.** It is not reflexive because $a < a$ is false. It is not symmetric because $1 < 2$ but $2 < 1$ is false. It is antisymmetric (the premise $a < b$ and $b < a$ is never true) and transitive by the usual order of integers. It is also irreflexive and asymmetric. **Question.** 3. Show that $R = \{(1, 1), (2, 2)\}$ on $A = \{1, 2\}$ is both symmetric and antisymmetric.

**Solution.** Every related pair is a loop. Reversing a loop gives the same pair, so symmetry holds. If $aRb$ and $bRa$, then necessarily $a = b$, so antisymmetry holds. **Question.** 4. Give a counterexample to transitivity for $R = \{(1, 2), (2, 3)\}$ on $\{1, 2, 3\}$.

**Solution.** $1R2$ and $2R3$ hold, but $1R3$ is false. Therefore $R$ is not transitive.

## 10 Common mistakes

Do not confuse symmetric with antisymmetric. For a universal relation property, use arbitrary elements; to disprove it, provide one concrete counterexample. In transitivity proofs, explicitly identify the intermediate element and the two assumed related pairs.